

9 The variation with distance r of the electric potential V of a charged object is shown in Fig. 9.1.

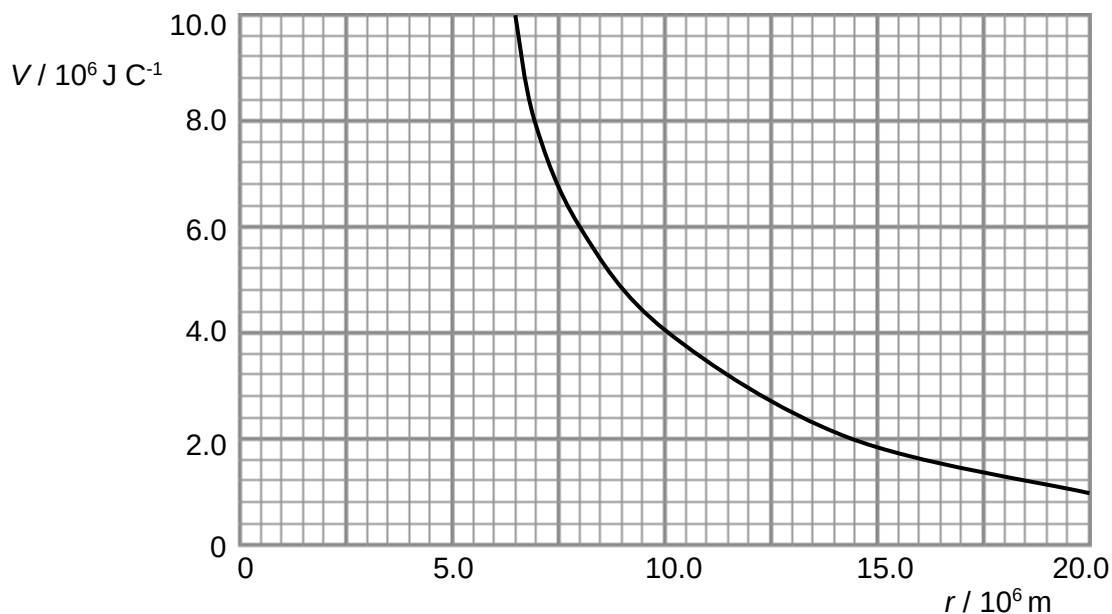


Fig. 9.1

- (a) The charged object is fixed in its position. A proton is initially at rest at 7.5×10^6 m from the centre of the charged object.
- Determine its kinetic energy when it has moved a distance of 7.0×10^6 m away from the charged object.

kinetic energy =	J	[3]
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L3	The proton would accelerate from 7.5×10^6 m to 14.5×10^6 m from the centre of the charged object.
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Electric potential at $7.5 \times 10^6 \text{ m} = 6.8 \times 10^6 \text{ J}$
Electric potential at $14.5 \times 10^6 \text{ m} = 2.0 \times 10^6 \text{ J}$

Gain in kinetic energy = Loss in electric potential energy

Final kinetic energy – 0 = $q \Delta V$

$$\text{Final kinetic energy} = (1.60 \times 10^{-19}) (6.8 \times 10^6 - 2.0 \times 10^6) \\ = 7.68 \times 10^{-13} \text{ J}$$

M1
A1

- (b)** On Fig. 9.2, draw a graph to show the variation with distance r of the electric field strength E for values of r from 7.5×10^6 m to 17.5×10^6 m.

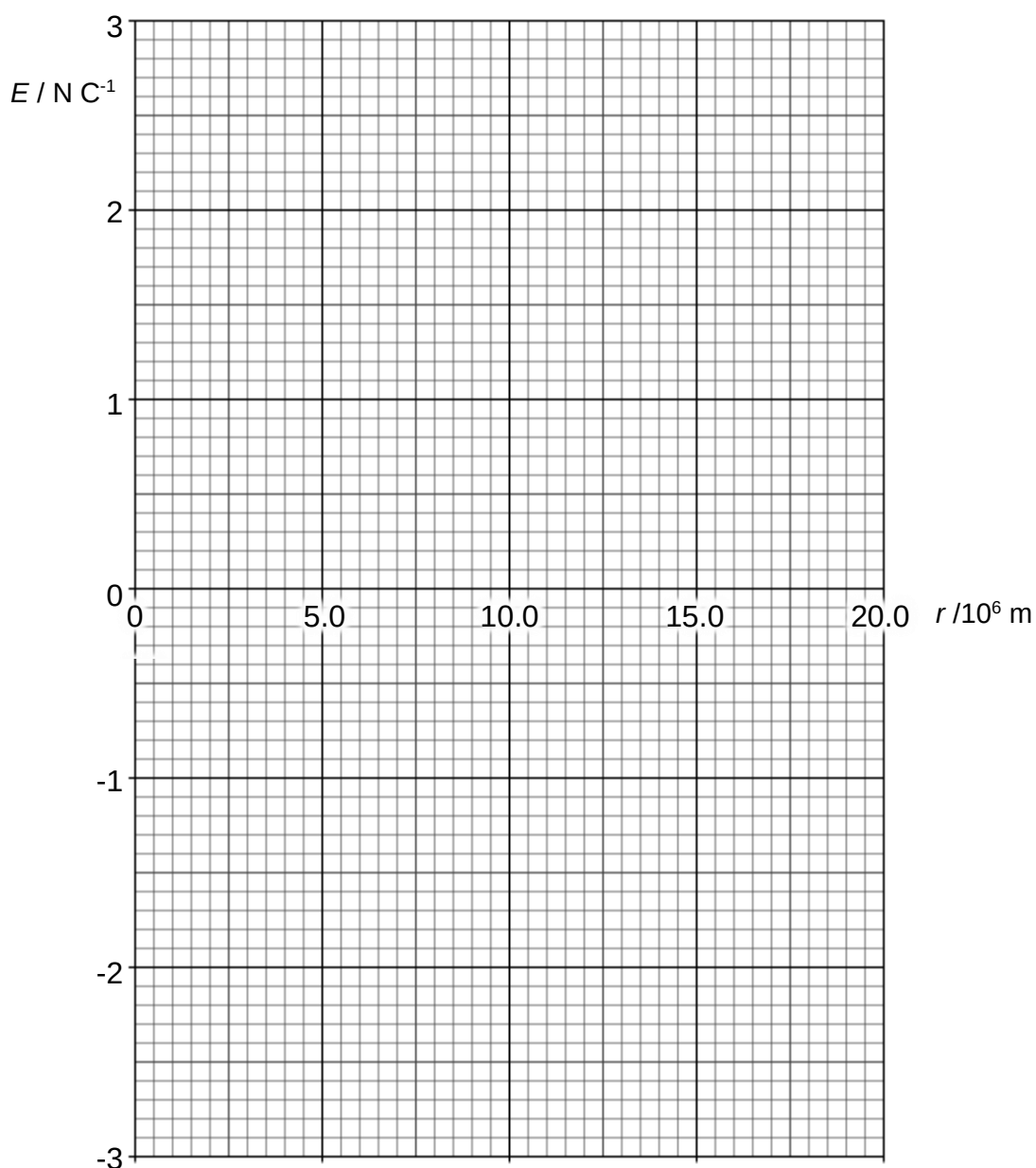


Fig. 9.2

[3]

L2 Electric field strength $E = -dV/dr$

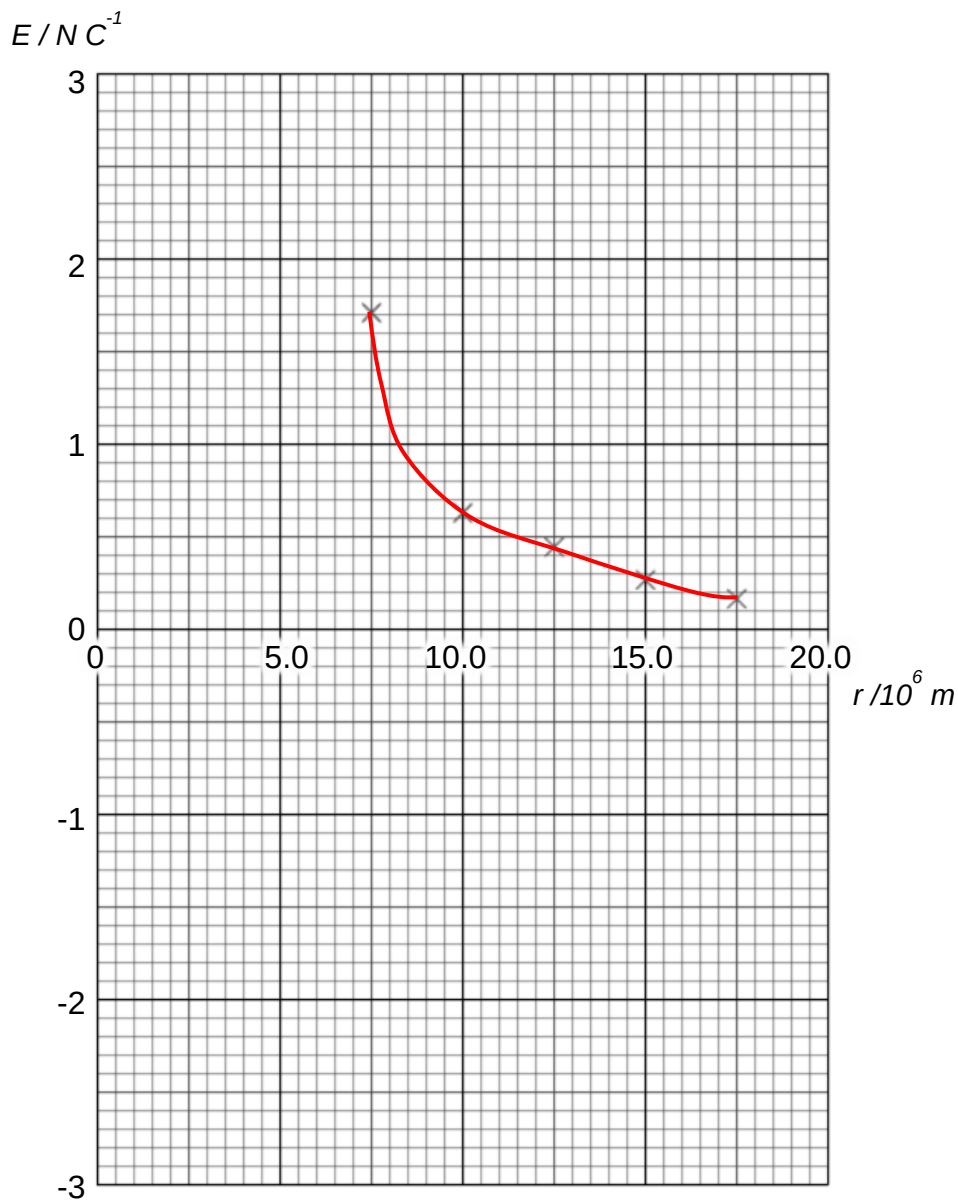
At $r = 7.5 \times 10^6 \text{ m}$,
 $dV/dr = \text{gradient of } V\text{-}r \text{ graph} = (2.4 - 8.4)/(10 - 6.5) = -1.71 \text{ V m}^{-1}$
 $E = +1.71 \text{ V m}^{-1}$

At $r = 10.0 \times 10^6 \text{ m}$,
 $dV/dr = \text{gradient of } V\text{-}r \text{ graph} = (0 - 8.8)/(16.5 - 2.5) = -0.629 \text{ V m}^{-1}$
 $E = +0.629 \text{ V m}^{-1}$

At $r = 12.5 \times 10^6 \text{ m}$,
 $dV/dr = \text{gradient of } V\text{-}r \text{ graph} = (1.2 - 7.2)/(16 - 2.5) = -0.444 \text{ V m}^{-1}$
 $E = +0.444 \text{ V m}^{-1}$

At $r = 15.0 \times 10^6 \text{ m}$,
 $dV/dr = \text{gradient of } V\text{-}r \text{ graph} = (0.8 - 5.6)/(19 - 1) = -0.267 \text{ V m}^{-1}$
 $E = +0.267 \text{ V m}^{-1}$

At $r = 17.5 \times 10^6 \text{ m}$,
 $dV/dr = \text{gradient of } V\text{-}r \text{ graph} = (1.2 - 4)/(18.5 - 1.5) = -0.165 \text{ V m}^{-1}$
 $E = +0.165 \text{ V m}^{-1}$



1 mark: at least 1 coordinate calculated correctly, showing understanding of $E = -dV/dx$.

B1

1 mark: at least 3 coordinates plotted to draw a curve in the positive region of the graph grid and in the required range of $7.5 \times 10^6 \text{ m}$ to $17.5 \times 10^6 \text{ m}$. (Allow for computation errors. However, the 3 coordinates should be well spread out over the correct range, e.g. at the 2 boundaries plus 1 coordinate midway between these 2 boundary coordinates.)

B1

1 mark: smooth curve and line of best fit drawn across the required range of $7.5 \times 10^6 \text{ m}$ to $17.5 \times 10^6 \text{ m}$.

B1

- (c) A certain planet has a radius of 1150 km. Fig. 9.3 below shows the variation with the distance x from the centre of this planet, of the gravitational potential ϕ near it. The planet may be assumed to be isolated in space.

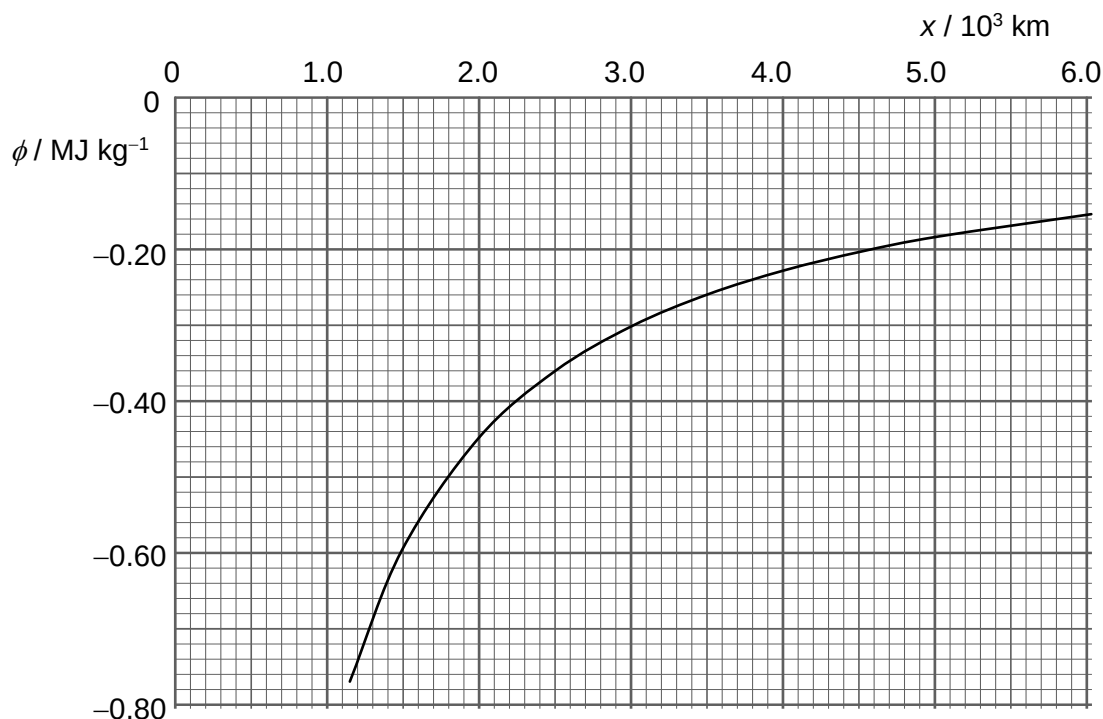


Fig. 9.3

- (i) Explain why gravitational potential has a negative value.

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[2]

- L1** Gravitational potential is defined to be zero at infinity.

Also, due to the attractive nature of gravitational force, to move a point mass from infinity towards another mass without a change in kinetic energy, an external agent would need to exert a force that acts opposite direction to the change in displacement of the mass being moved, thus work done per unit mass by the external agent is negative.

Thus gravitational potential at any point other than at infinity is negative.

- Relate back to definition of gravitational potential: Gravitational potential at a point is the work done per unit mass by an external agent in bringing a point mass from infinity to that point (without a change in kinetic energy).
- 'potential' is a 'per unit mass' quantity, hence reference is made to work done 'per unit mass' in the explanation.

B1

B1

		(ii)	Use Fig. 9.3 to determine the mass of the planet.	
			mass = kg	[2]
		L2	Using $x = 3.0 \times 10^3$ km with its corresponding potential $\phi = -0.30 \times 10^6$ J (or any other coordinates), $\phi = -\frac{GM}{x}$ $-0.30 \times 10^6 = -\frac{(6.67 \times 10^{-11})(M)}{3.0 \times 10^3 \times 10^3}$ $M = 1.3493 \times 10^{22} = 1.35 \times 10^{22} \text{ kg}$	M1 A1
		(iii)	A moon of the planet has a circular orbit of radius 3.0×10^3 km. The period of its orbit is 3.44×10^4 s. Calculate the centripetal acceleration of the moon.	
			centripetal acceleration = m s^{-2}	[2]
		L2	$a = r\omega^2 = r\left(\frac{2\pi}{T}\right)^2 = 3.0 \times 10^3 \times 10^3 \left(\frac{2\pi}{3.44 \times 10^4}\right)^2$ $a = 0.10008 = 0.100 \text{ m s}^{-2}$	M1 A1
		(iv)	Explain why the gravitational field strength at the position of the moon has the same magnitude and same direction as the centripetal acceleration of the moon.	
				
				
				
				
				
				
				
				[3]
		L2	Gravitational field strength g is defined as the gravitational force F_g per unit mass, hence $F_g = \text{mass of moon} \times g$. By Newton's second law of motion, centripetal force $F_c = \text{mass of moon} \times \text{centripetal acceleration } a_c$. As gravitational force F_g provides the centripetal force F_c for the moon to orbit the planet, g is of the same magnitude and same direction as a_c.	B1 B1 B1
		(v)	The mass of the moon is 1.52×10^{21} kg. Calculate the total energy of the moon.	

			total energy = J	[3]
		L2	Gravitational potential energy $= -\frac{GMm}{r}$ $= -\frac{(6.67 \times 10^{-11})(1.3493 \times 10^{22})(1.52 \times 10^{21})}{3.0 \times 10^3 \times 10^3}$ $= -4.5599 \times 10^{26} \text{ J}$ Kinetic energy $= \frac{1}{2}mv^2 = \frac{1}{2}(1.52 \times 10^{21})(r\omega)^2$ $= \frac{1}{2}(1.52 \times 10^{21})\left((3.0 \times 10^3 \times 10^3)\left(\frac{2\pi}{3.44 \times 10^4}\right)\right)^2$ $= 2.28191 \times 10^{26} \text{ J}$ Total energy = Gravitational potential energy + Kinetic energy $= -4.5599 \times 10^{26} + 2.28191 \times 10^{26} = -2.28 \times 10^{26} \text{ J}$ OR Derive to show that $KE = \frac{1}{2} \frac{GMm}{r}$ Total energy = KE + GPE = $\frac{1}{2} \frac{GMm}{r} + \left(-\frac{GMm}{r}\right)$ $= -\frac{1}{2} \frac{GMm}{r}$ $= -\frac{1}{2} \frac{(6.67 \times 10^{-11})(1.3493 \times 10^{22})(1.52 \times 10^{21})}{3.0 \times 10^3 \times 10^3}$ $= -2.27995 \times 10^{26} \text{ J}$ $= -2.28 \times 10^{26} \text{ J}$	M1 M1 A1
	(d)	State and explain one similarity and one difference in the variations in the electric potential and gravitational potential shown in Fig. 9.1 and Fig. 9.3 respectively.		
		similarity:		
				
				
		difference:		
				
				[2]
	L1	Similarity: The magnitudes of both electric potential and gravitational potential are inversely proportional to the distance from the centre of the object. Difference: Gravitational potential is always negative for any mass, whereas electric potential is positive for a positive charge (and negative for a negative charge).		B1 B1

[Total: 20]

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